DATA SCIENCE PROJECT

Penguins

PURPOSE

- The goal of this project was to perform a full cycle of the data science process on the 'penguins' dataset.
- Exploratory Data Analysis (EDA) was conducted, followed by model construction.
- Insights were then gained from both the EDA phase and the model construction phase.

EXPLORATORY DATA ANALYSIS (EDA)

- Data discovery was first conducted to understand the structure, variables and values.
- Data cleaning was then performed to improve the quality of the dataset.
- Data validation assessed the dataset to ensure consistency and balance.
- Data structured transformed the dataset into a more useful form.
- Data visualisations provided insight on attributes and relationships.

MODEL CONSTRUCTION AND EVALUATION

- A Simple Linear Regression model was built to predict body mass based on flipper length. The assumptions of the model were evaluated to ensure a valid model.
- A K-Means clustering model was constructed to organise the data into distinct groups.
- A Random Forest Classification Model was created to predict the sex of the penguin given a series of variables.
- The models were optimized to improve performance.
- The models were then evaluated to determine performance.

EDA INSIGHTS

The following insights were gained from the exploratory data analysis:

- The 'Gentoo' species weighs just over 1 kg more than the other two species.
- Males are around half a kilogram heavier than females.
- There is a strong positive correlation between flipper length and body mass.
- Males have longer bills, longer flippers, deeper bills and more body mass.
- Penguins with deeper bills have shorter flipper and less body mass.

MODEL INSIGHTS

The following insights were gained from development of the models:

- The Simple Linear Regression model was reasonably good at predicting body mass given flipper length.
- The Clustering model was not perfect but not too bad, especially after refinement.
- The Random Forest model had a score of 92.9% over accuracy, precision, recall and F1-Score.
- Body mass and bill depth were the most important features for the Random Forest Model, with bill length and flipper length contributing aswell.